LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 4

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO4

Paper: Sample Multiple Choice Questions - ANSWER KEY

Maximum Marks: 40 x 1 = 40

Unit: 4

Unit Title: Dimensionality Reduction and Representation Learning

Unit Mapped RBT: L5 (Evaluate)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the matrix whose eigenvectors give the principal components of a dataset.

A) The covariance matrix <-- correct
B) The proximity matrix
C) The confusion matrix
D) The adjacency matrix

Answer: A. PCA finds the eigenvectors of the covariance matrix of the centred data; those directions are the components, in order of the variance they carry.

Q2. Identify what an eigenvalue of the covariance matrix reports about its component.

A) The number of records in that direction
B) The variance captured along that direction <-- correct
C) The angle between two components
D) The reconstruction error of the component

Answer: B. The eigenvalue is the variance along its eigenvector, which is why the components can be ranked simply by ordering the eigenvalues.

Q3. State the part of an autoencoder that maps the input down to the bottleneck layer.

A) The decoder
B) The classifier
C) The encoder <-- correct
D) The optimiser

Answer: C. The encoder compresses the input to the bottleneck; the decoder attempts to rebuild the original from that compressed representation.

Q4. Name the quantity an autoencoder is trained to minimise.

A) The classification accuracy
B) The explained variance
C) The silhouette score
D) The reconstruction error <-- correct

Answer: D. The network is trained to reproduce its own input, so the loss is the difference between the input and the output the decoder produces.

Q5. Identify the t-SNE setting that governs how many neighbours each point is balanced against.

A) Learning rate
B) Perplexity <-- correct
C) Batch size
D) Epsilon

Answer: B. Perplexity acts as a soft count of effective neighbours. Changing it can visibly change the layout, which is why it must always be reported.

Q6. State the largest number of principal components that a dataset of p features can produce.

A) p - 1
B) 2p

C) Two

D) p <-- correct

Answer: D. The components form a new basis for the same space, so there can be at most as many of them as there are original features.

Q7. Explain why the data must be centred before the principal components are computed.

A) Variance is defined as spread about the mean <-- correct

B) Centring removes the outliers from the table

C) Centring makes every feature take the same units

D) Centring guarantees the components are orthogonal

Answer: A. If the data is not shifted to the origin, the first component points towards the mean rather than along the direction of greatest spread.

Q8. Distinguish an undercomplete autoencoder from an overcomplete one.

A) Undercomplete has a bottleneck wider than the input

B) Undercomplete uses no decoder at any stage

C) Undercomplete has a bottleneck narrower than the input <-- correct

D) Undercomplete is trained on labelled data only

Answer: C. It is the narrow bottleneck that forces a compressed representation. An overcomplete network can otherwise simply copy the input straight through.

Q9. Describe what the manifold hypothesis claims about high-dimensional data.

A) It fills the whole of the high-dimensional space

B) It always contains exactly two natural clusters

C) It lies near a surface of far fewer dimensions <-- correct

D) It cannot be reduced without losing everything

Answer: C. Real data is usually constrained by the process that produced it, so the records occupy a thin curved sheet rather than the whole space. This is what makes reduction possible.

Q10. Interpret an explained variance ratio of 0.62 reported for the first principal component.

A) It captures 62 per cent of the total variance <-- correct

B) It is correlated 0.62 with the second component

C) It reconstructs 62 per cent of the records exactly

D) It requires 62 per cent of the original features

Answer: A. The ratio is that component's eigenvalue divided by the sum of all of them, so it states the share of overall spread that lies along that direction.

Q11. Predict what happens to a t-SNE layout when the perplexity setting is changed.

A) The layout is completely unaffected by it

B) The records are removed from the plot entirely

C) The axes acquire meaningful units of measure

D) The apparent number and shape of the groups can change <-- correct

Answer: D. Perplexity decides the neighbourhood scale the method tries to preserve, so different values can produce visibly different pictures of the same data.

Q12. Recognize why t-SNE cannot place a new record onto an existing plot without being refitted.

A) It discards the original features after fitting

B) It learns a layout, not a reusable mapping <-- correct

C) It requires the record to carry a class label

D) It can only handle exactly two dimensions

Answer: B. t-SNE optimises the positions of the points it was given. PCA, by contrast, produces a projection that can be applied to any new record.

Q13. Explain the sense in which Principal Component Analysis is an unsupervised technique.

A) It requires a label for every record used

B) It predicts a numeric target for each record

C) It measures accuracy against known answers

D) It uses the features alone and never a target <-- correct

Answer: D. PCA looks only for the directions of greatest variance in the features. It has no notion of which direction is useful for any particular prediction.

Q14. Describe what a denoising autoencoder is trained to do.

A) Rebuild a record from its class label alone

B) Remove every outlier from the training table

C) Rebuild a clean record from a corrupted copy <-- correct

D) Compress the data without any decoder at all

Answer: C. Noise is added to the input while the target stays clean, which forces the bottleneck to hold the underlying structure rather than the noise.

Q15. Explain why a linear autoencoder trained with a squared-error loss recovers the same subspace as PCA.

A) Both are trained on the class labels of the data

B) Both find the best linear subspace for reconstruction <-- correct

C) Both require the features to be uncorrelated first

D) Both retain every one of the original features

Answer: B. With no non-linearity the encoder is a linear projection, and the projection minimising squared reconstruction error is exactly the leading principal subspace.

Q16. Predict the effect of a single extreme outlier on the first principal component.

A) It can pull the component towards itself <-- correct

B) It is automatically excluded by the method

C) It has no effect because of the centring step

D) It causes the component to become undefined

Answer: A. PCA maximises variance, and a distant point contributes enormously to the variance along the direction pointing at it. Robust variants exist for this reason.

Q17. Solve the following: five components explain 40, 25, 15, 12 and 8 per cent of the variance. Determine the smallest number retaining at least 80 per cent.

A) 2

B) 3 <-- correct

C) 4

D) 5

Answer: B. The cumulative shares are 40, 65 and 80 per cent, so three components are enough to reach the threshold.

Q18. Compute the cumulative variance explained by the first two of four components explaining 55, 20, 15 and 10 per cent.

A) 75 per cent <-- correct

B) 55 per cent

C) 20 per cent

D) 90 per cent

Answer: A. The cumulative figure is simply the running total: 55 + 20 = 75 per cent.

Q19. Apply the Kaiser criterion to the eigenvalues 3.2, 1.8, 1.1, 0.7 and 0.2, and determine how many components are retained.

A) 2

B) 4

C) 5

D) 3 <-- correct

Answer: D. The Kaiser rule keeps components whose eigenvalue exceeds 1, which here means the first three.

Q20. Solve the following: a dataset of 500 features is reduced to 25 components. Determine the reduced size as a fraction of the original.

A) One fifth

B) One tenth

C) One twentieth <-- correct

D) One fiftieth

Answer: C. 25 / 500 = 1 / 20, so the representation is one twentieth of its original width.

Q21. Use the definition of reconstruction error to determine what a value of exactly zero would mean.

A) The decoder reproduces every input exactly <-- correct

B) The bottleneck holds no information at all

C) The network has not yet been trained once

D) The inputs were all identical to each other

Answer: A. Zero error means the output matches the input for every record, so the bottleneck retains everything needed to rebuild them.

Q22. Compute the variance explained by the fourth component when the first three together explain 61 per cent and the first four explain 69 per cent.

A) 4 per cent

B) 61 per cent

C) 8 per cent <-- correct

D) 69 per cent

Answer: C. The difference between the two cumulative figures is that component's own share: 69 - 61 = 8 per cent.

Q23. Apply Principal Component Analysis to a dataset of 12 features and determine how many components are produced when no limit is set.

A) 11

B) 12 <-- correct

C) 6

D) 2

Answer: B. Without a limit, PCA returns as many components as there are features; the choice of how many to keep is made afterwards.

Q24. Examine the following and determine the consequence: Principal Component Analysis is applied to raw columns holding height in metres alongside salary in rupees.

A) The height column dominates the first component

B) The two columns contribute equally to the result

C) The components come out perfectly uncorrelated

D) The salary column dominates the first component <-- correct

Answer: D. Variance depends on the unit, and salary in rupees has a vastly larger numeric spread. The components then describe the salary column and little else.

Q25. Use the following explained variances to determine how many components reach a 95 per cent threshold: 70, 18, 7, 3 and 2 per cent.

A) 2

B) 4

C) 3 <-- correct

D) 5

Answer: C. The cumulative shares are 70, 88 and 95 per cent, so three components meet the threshold exactly.

Q26. Compare Principal Component Analysis with t-SNE on the ability to transform a record that was not in the training set.

A) t-SNE can transform it, PCA cannot

B) Both of the two can transform it directly

C) Neither of the two can transform it at all

D) PCA can transform it, t-SNE cannot <-- correct

Answer: D. PCA yields a fixed projection that applies to any record. t-SNE only positions the points it was fitted on, so a new record requires a refit.

Q27. Evaluate the claim that a principal component explaining very little variance can always be discarded safely.

A) It is unsafe, since low variance is not low usefulness <-- correct

B) It is always safe, as variance measures usefulness

C) It is always safe, since components are uncorrelated

D) It is unsafe only when the data is standardised

Answer: A. PCA ranks directions by spread, not by relevance. A small, consistent signal may sit in a low-variance direction and be thrown away with it.

Q28. Assess the reading of a t-SNE plot in which the gap between two islands is taken as a measure of how different the two groups are.

A) The reading is sound, as t-SNE preserves all distances

B) The reading is unsound, as global distances are not preserved <-- correct

C) The reading is sound provided the data was scaled

D) The reading is unsound only for very small datasets

Answer: B. t-SNE optimises local neighbourhoods. The space between well-separated islands carries no reliable meaning, so the gap must not be quantified.

Q29. Justify the use of Principal Component Analysis before clustering a table of 800 heavily correlated features.

A) It supplies the number of clusters to be used

B) It guarantees that the clusters will be spherical

C) It replaces the need to validate the clustering

D) It removes redundancy and restores contrast in the distances <-- correct

Answer: D. Correlated features count the same information many times and flatten the distances. Projecting onto a few uncorrelated directions restores usable structure.

Q30. Determine the more suitable approach when every reduced feature must be traceable to a quantity that was actually measured.

A) Principal Component Analysis

B) Feature selection <-- correct

C) An autoencoder bottleneck

D) A t-SNE embedding

Answer: B. Selection keeps original columns, each carrying its own meaning. Any extraction method produces mixtures that cannot be described in the source terms.

Q31. Recommend the correct response to a scree plot that falls smoothly with no visible elbow.

A) Retain exactly two components in every case

B) Abandon dimensionality reduction altogether

C) Choose by a variance threshold and report the choice <-- correct

D) Retain every component the method produced

Answer: C. When the curve gives no bend to read, the decision has to rest on an explicit rule, and the rule used must be stated so a reader can judge it.

Q32. Judge whether a very low reconstruction error proves that an autoencoder representation will serve the task that follows.

A) It does not, since faithful is not the same as useful <-- correct

B) It does, because the information is provably intact

C) It does, provided the bottleneck is narrow enough

D) It does not, because the error is always unreliable

Answer: A. A low error shows the code can rebuild the input. It says nothing about whether the code separates the groups or supports the decision the project needs.

Q33. Measure the share of the original information a Principal Component Analysis projection retains by selecting the appropriate quantity.

A) The cumulative explained variance ratio <-- correct

B) The silhouette coefficient of the projection

C) The number of components produced

D) The Davies-Bouldin Index of the result

Answer: A. The cumulative ratio states what fraction of the total variance the retained components account for, which is the standard summary of what survives.

Q34. Assess the risk of applying Principal Component Analysis when the informative signal has a small variance.

A) The signal is always placed in the first component

B) The method refuses to run in that situation

C) The signal is amplified by the projection step

D) The signal may be dropped with the discarded components <-- correct

Answer: D. Components are ranked by variance alone, so a faint but meaningful pattern can end up far down the list and be discarded as noise.

Q35. Evaluate the claim that UMAP should always be preferred to t-SNE.

A) UMAP is always better because it is faster

B) Neither is always better; the choice depends on the task <-- correct

C) t-SNE is always better because it is older

D) The two produce identical layouts in every case

Answer: B. UMAP is generally quicker and keeps more of the broad arrangement, but both are stochastic and neither gives distances that can be measured. The purpose decides.

Q36. Recommend what must accompany a two-dimensional embedding placed in a report.

A) The number of clusters the analyst expected

B) The accuracy the embedding achieved on test data

C) The method, its settings and the random seed used <-- correct

D) Nothing further, since the picture speaks for itself

Answer: C. Without the method and its settings the picture cannot be reproduced or challenged, and a different seed may well have produced a different arrangement.

Q37. Determine the representation to prefer when the finished model must run on a device with very little memory.

A) Every one of the original raw features

B) A two-dimensional t-SNE layout of the data

C) A small number of components or a narrow bottleneck <-- correct

D) The full proximity matrix of the records

Answer: C. A compact fixed projection can be applied cheaply on the device, whereas raw features cost memory and a t-SNE layout cannot be applied to new input at all.

Q38. Critique a project that reports 99 per cent of the variance retained while the clustering that followed became noticeably worse.

A) The variance figure must have been miscalculated

B) Retained variance is not evidence of retained structure <-- correct

C) Clustering always degrades after any reduction

D) The project should have retained 100 per cent

Answer: B. Almost all the spread can survive while the particular directions that separated the groups are lost. The result must be judged on the clustering, not the variance.

Q39. Judge the claim that dimensionality reduction improves the performance of any method it is applied to.

A) True, because noise is always removed by it

B) True, because fewer features always help

C) False, because it never changes performance

D) False, since useful information can be discarded too <-- correct

Answer: D. Reduction helps when the features are redundant or noisy. When each carries distinct signal, compressing them simply throws part of that signal away.

Q40. Evaluate the decision to train an autoencoder rather than run Principal Component Analysis on a dataset holding only 300 records.

A) PCA is the better choice, as the network would overfit <-- correct

B) The autoencoder is better, as it is always stronger

C) The two would give identical results at this size

D) Neither method can be applied to 300 records

Answer: A. An autoencoder has many parameters and needs a large sample to learn a general mapping. On 300 records it memorises rather than generalises.

----- End of Sample Question Set -----